

## Determining the Age of Open Cluster M67 via BV Photometry and Isochrone Fitting

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### ABSTRACT

In this lab, we utilize a ground based 14.5 inch telescope along with a B and V broad-band filter at Mt. Stony Brook observatory to perform BV photometry on open cluster M67 (NGC 2682). From these images we measure the brightness and color of several cluster stars and build a color–magnitude diagram (CMD), a plot of stellar brightness versus color, which we then compare to computer generated stellar isochrones from the PARSEC stellar evolution library (Bressan et al. (2012)) to predict the cluster’s age. Through our analysis we were able to successfully identify a lower bound of 1 Gyr but struggled to differentiate between upper bounds and support the expected literature value of 4 Gyr (Richer et al. (1998)) due to lack of stars in the main sequence turnoff point.

*Keywords:* Open clusters, M67, NGC 2682, BV photometry, Aperture photometry, Isochrone, Isochrone fitting

### 1. INTRODUCTION

One of the most fundamental aspects of astrophysics is understanding the behavior and composition of stars and their change over their lifetime. The evolution of a star is most significantly impacted by two major properties, it’s mass and it’s age. More massive stars are capable of emitting more energy per second achieving brighter luminosities, and in turn, burn fast and live short lifetimes. Less massive stars have lower luminosities; thus, they preserve more energy over time and have much longer lifespans. Due to the effect of a star’s initial mass on the production of it’s energy, each star goes through certain evolutionary changes that impact it’s corresponding luminosity and temperature. To analyze and understand stellar evolution, astronomers take advantage of this fact and utilize a Color Magnitude Diagram (also known as a Hertzsprung-Russell Diagram). A CMD is a scatter plot comparing the magnitude (brightness) of stars/galaxies against their color (temperature). This diagram can highlight and classify various stages in evolution. These classifications include (but are not limited to) the main sequence, where core hydrogen burning occurs and most stars typically spend most of their lifetime; the red giant, horizontal, and asymptotic giant branches where various stages of helium burning occurs; the supergiants, where the most massive and luminous stars exist; and the low-luminosity white dwarfs where stars that are of sufficiently low mass reach at the end of their life. Since we know that stars evolve differently based on their mass, we know that stars of the same age can exist at different regions of the CMD.

While it may be difficult to study and understand stellar evolution through the analysis of a single star, open clusters provide great opportunities for a multitude of reasons. Open clusters are composed of tens to thousands of loosely bound stars that are formed from the same giant molecular cloud. Molecular clouds are relatively compact regions of interstellar gas and dust left over from galaxy formation. Within these clouds are high-density regions known as clumps, that collapse under sufficient gravitational force, allowing for star formation. These open clusters that form stars are often referred to as “stellar nurseries”. Due to their shared origins stemming from the same molecular cloud composed of the same material, stars from open clusters have similar age, distance, and chemical composition. This makes open clusters perfect for studying stellar evolution as they constrain the amount of variable parameters between them and permit us to physically observe how initial mass impacts stars.

Despite this, difficulties still exist in studying stellar evolution because the timescales at which stars evolve are far too vast to be observed. Even among the stars with the shortest lifespans, they evolve over the course of millions of years. This cannot feasibly be observed over the course of human history. In order to actually date these open clusters, we utilize stellar isochrones. Isochrones are a curve on the CMD that represents a population of stars that all share the same age but vary in mass. These curves are theoretical models that are computationally generated based off of various core principles in astrophysics that align with previous observations. By generating isochrones of various ages, we can utilize them to date open clusters by plotting the magnitude and colors of their stars from these clusters on the CMD diagram and seeing how they align between isochrones of various ages. This allows us to potentially find upper and lower bounds for the possible age of a given cluster.

In this lab, we image the open cluster M67 (NGC 2682) in two filters (B and V) using the Mt. Stony Brook 14-inch telescope. From these images we measure the brightness and color of hundreds of cluster stars and construct the color magnitude diagram (CMD). We then compare the shape of the corresponding CMD to a series of computer-generated stellar isochrones from the PARSEC stellar evolution library (Bressan et al. (2012)). These isochrones span across various ages to predict the age of M67. M67 was an ideal target for the purposes of this lab because its stars were at a convenient brightnesses for our telescope ( $V \approx 10-16$ ), it is well-placed in the sky, and it should produce a clear, distinctive CMD shape that makes age determination straightforward. Additionally, M67 is a well studied and document open cluster, meaning that we have existing literature values for many of the assumptions we make along this process including it's age and metallicity. Our quantitative goal was to determine the cluster age to within  $\pm 0.5$  Gyr and compare our result to the accepted literature value of 4 Gyr (Richer et al. (1998)). Furthermore, by measuring these values for M67, we can potentially confirm the accuracy of several of these known literature values and the accuracy of the isochrone generation from the PARSEC stellar evolution library.

In this report, we will first discuss the data acquisition process in Section 2. This includes details on the observation, setting, time, conditions, and equipment. Section 3 covers data reduction, and analysis. This includes the steps required for adjusting for dark current, creating flat fields, aligning the science frames, magnitude calibration, and adjusting for absolute magnitudes. At the end of section 3 and the start of section 4, we discuss the results and the resulting CMD diagrams with our captured stars from M67 displayed. Finally, in section 5, we discuss our results including the discovery of a lower bound, a summary what factors may have impacted our results, and how it can be improved in the future.

## 2. DATA ACQUISITION

The observations and data for this lab were acquired on the evening of Wednesday, April 8th, 2026 at Mt. Stony Brook observatory (40.91476°N 73.12568°W) between the hours of 20:00 and 22:00. The observations were initially scheduled for April 6th but were postponed due to cloud coverage that prevented the acquisition of high quality data. The weather conditions on the day of observations were ideal with no visible cloud coverage, more than sufficient to acquire usable data for the purposes of the lab. The observations were conducted by the coauthors, 114956855 and 115541173, under the supervision of the course TA, Béla Arwen. The author was unable to attend and assist with the observations due to unexpected injury, but was able to video-call in and receive updates from the collaborators.

In our observation process, we utilized the 14 Inch Meade LX200-ACF telescope connected to a Mesu-200 equatorial mount. To shift to our targets, we utilized Cartes du Ciel, a software with GoTo functionality, and the mount's associated manual hand-pad for minor adjustments to get targets into frame. Additionally, we mounted a ZWO ASI2600MM DUO camera to the back end of the telescope to capture the data and save it to an external laptop. Of it's set of broad band BVRI filters, we utilized the B and V filter, denoting "blue" and "visible", respectively, to capture the magnitude at the two distinct wavelength ranges and calculate the color value of each star. We optimized storage by setting the binning factor to 4x4 as the common existing light pollution in the local environment would likely offset any useful additional data from reducing the binning factor.

Once the ZWO ASI2600MM DUO camera was successfully mounted to the telescope, we began with our observations of our main target, a portion of the open cluster M67 (NGC 2682). Using the Cartes De Ciel software, we slewed to NGC 2682 and then manually adjusted the telescope's view using the mount's hand-pad until the previewed stars matched the positioning of the stars in M67's star chart. The first observations were captured with the V filter at a 45 second exposure time. 15 observations were taken in the V filter in total. This process was then repeated upon swapping to use the B filter instead and a series of 15 exposures at the same exposure time were also captured. The

exposure time of 45 seconds maximized the detected photons from M67 in each filter while largely avoiding blurring due to their change in position across the sky because of Earth’s rotation.

We then slewed to the region SA 101 and repeated this process. Once stars were located in the preview, a series of 10 45 second exposures were captured in both the V and B filter. This portion of the observation was ultimately deemed unnecessary following the observations and will be discussed further in the Data Reduction and Analysis section.

Following the science images of M67 and the standard star field, SA101, a series of dome flats was taken to adjust for variations in sensitivity between pixels during calibration. This was accomplished by taking a series of exposures with the telescope pointed at our dome with the primary lights on. The uniformity of the dome and the short distance allows for each pixel to receive approximately the same amount of uniform light, effectively making a high-quality flat frame. We took a series of 5 flat frames for both the B and V filter, with an exposure time of 17 and 4 seconds, respectively.

Lastly, a series of dark frames were taken at each exposure time that was used among all of the science images and the flat frames, 4, 17, and 45 seconds. The possibility of any extraneous photons being detected by the sensor were limited through closing the dome, covering the telescope with the lens cap, and shutting off any interior light in Mt. Stony Brook Observatory.

### 3. DATA REDUCTION AND ANALYSIS

#### 3.1. *Adjusting for Dark Current*

Before any of the data from our science frames can be analyzed, each of the frames must first be corrected for dark current, spontaneous electrons that may stem from thermal noise making the detection of a photon. The frames must also be corrected for hot pixels, pixels with large dark current that over-count even in short or no exposure time. This is achieved through our dark frames, exposures taken with a closed shutter that minimize the possibility of any real photon detection that were taken for every exposure time used in all our other frames. We must first create the master dark frame for every captured exposure time: 4, 17, and 45 seconds. This is done by median combining each dark frame in a given exposure time, such that we get an estimate for the potential dark current in our science frame that is consistent with that length of exposure.

Next, we subtract the appropriate master dark from each science, reference star, and flat-field frame.

#### 3.2. *Creating and applying Flat Fields*

Once the respective master dark frame is subtracted from each flat-field frame, the flat-field frames of a given exposure time must be median combined. Median-combining those given frames for each filter minimizes the effects of any cosmic rays, random fluctuations in our detections, and noise from photon counting. From there, the master flat-field for each filter is created by normalizing the median-combined frame by it’s typical count level. This is achieved by dividing the combined frame by it’s mean counts.

Once the master flat-frames are constructed, they are applied to each of the dark-subtracted science frames. Typically, to apply the master-flat frames, we would divide each of the science images by the respective master flat frame. While this is still the approach we ultimately take, we faced the unexpected issue of our flat field frames being composed of slightly different dimensions than the rest of our images between the science and dark frames. The flat field dimensions and the dimensions of every other frame diverge: Our science and dark frames had dimensions of (x=392, y=262) and our flat frames had dimension of (x=390, y=261), 2 x 1 pixel smaller. In order to address this issue, we crop each of the science images after dark subtraction so that they align. The positioning of the crop was decided by manually matching a warm pixel in the dark frames with the corresponding warm pixel in the flat field. A warm pixel was located at x = 144 and y = 99 in the dark frame and at x= 142 and y=99 in the flat field frame. Thus, because the warm pixel came two pixels earlier in the x-axis, the first two columns of the science frames must be removed. Following this line of reasoning, because the y position of the warm pixel does not change between frames, the flatfield must be missing the last row (y=262), so the last row of the science frames must be cropped out. Following cropping each of the science frames, we successfully divide each of them by the corresponding master flat-field for each filter and obtain the sensitivity corrected science frames.

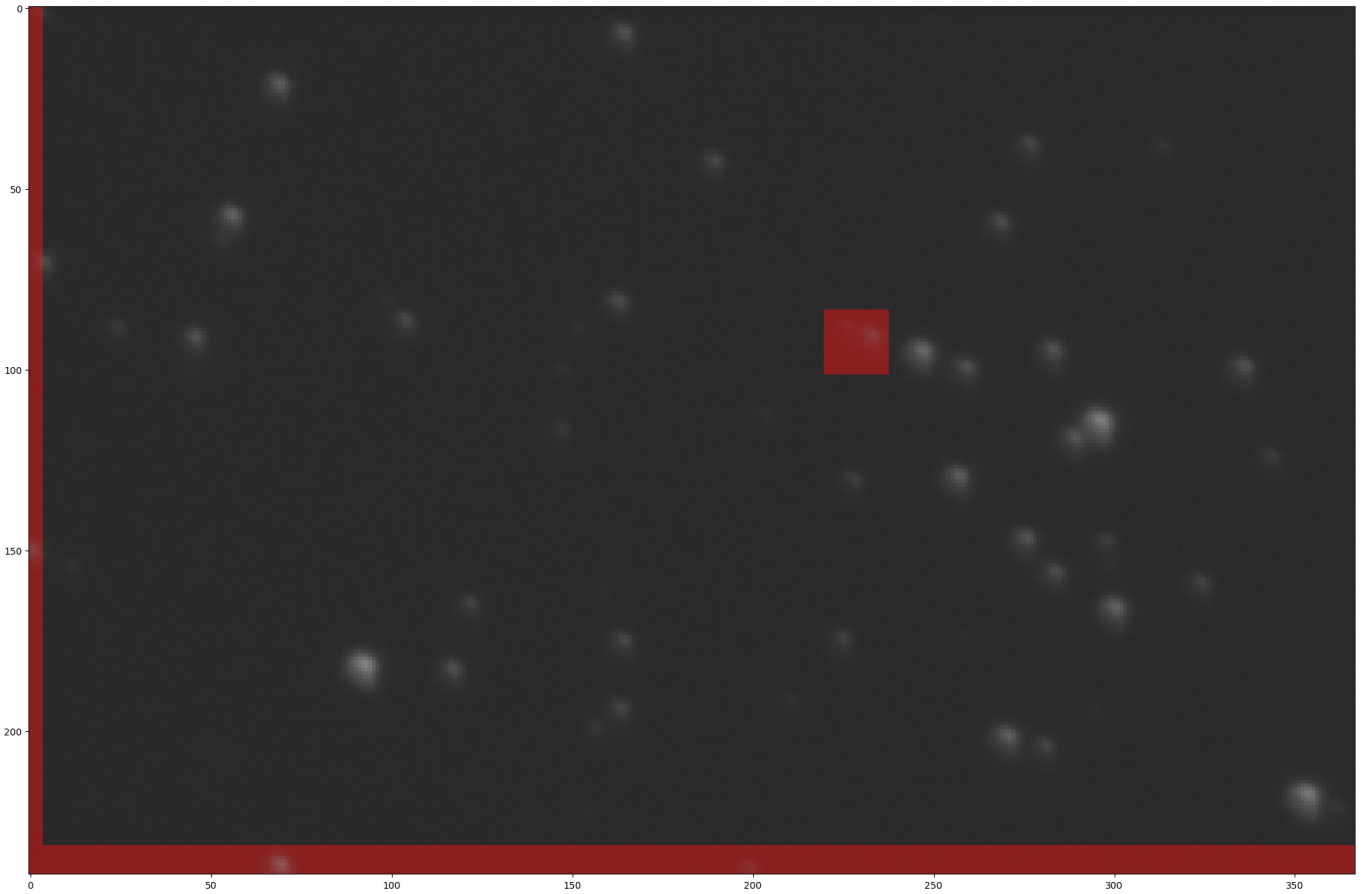
#### 3.3. *Aligning and Combining the Science Frames*

In order to median combine each of the dark subtracted and sensitivity corrected science frames of M67 and SA101, each of the frames in each filter must be aligned. This is necessary as the position of our mount and telescope remains

still throughout the observation process as the Earth rotates, shifting the position of these stars in our frame. This is accomplished by choosing the first frame in a given collection to be the "reference" that every other frame is aligned to. From there, we use the `phase_cross_correlation` function from the `skimage` python package to translate and shift each frame such that they are aligned. This process can also be done manually, but the use of cross correlation from the result from this existing package saves time and is then manually reviewed and adjusted until the index of any given star in each frame is consistent. Once this is accomplished for the B and the V filter frames, the aligned frames can be median-combined. From there, we repeat this process on the median-combined B frame and V frame. After those two frames are aligned, we crop them, ensuring that only rows that contain flux in both frames are included. This produces our final B and V flux frame.

### 3.4. Detecting Stars

Our next step involved finding the x and y coordinate of the center of each star in our final science images. We utilized the `photutils` library, a Python library that provides tools functionality for detecting and performing photometry of astronomical sources. To detect the pixel coordinates of each of the star, we used `DAOStarFinder` from `photutils` which utilizes the DAOFIND algorithm (Stetson (1987)). The DAOFIND algorithm searches for local density maxima that have a peak amplitude above a specified threshold. For the function, we set the threshold value required equal to 5 times the standard deviation of the counts in the final B frame. Before we run the algorithm we define mask regions where no stars will be detected. This is necessary for edge-cases such as stars that are along the edge whose pixels are cut off by the borders of the frame as well as overlapping stars who are too close to be completely and distinctly resolved in our capture of their flux and will skew their actual magnitude.



**Figure 1.** Preview of the final B filter image along with the masked regions for peak detection.

From the output of this function, we extract the x and y centroids from the list and convert them into integers to get the corresponding x and y coordinates for each of the stars in our observations. Following masking the ineligible stars and running the function, we were able to successfully detect 43 stars

### 3.5. Aperture Photometry and Extracting Flux from Targets

From the obtained x and y coordinates of each of the star centers, we can utilize aperture photometry to calculate the instrumental magnitude of the stars within M67. In aperture photometry, we sum the observed flux within a certain radius of the center of the object. Then we subtract the total contribution of the sky background flux within the same region.

$$N_{target} = N_{total} - N_{sky} \quad (1)$$

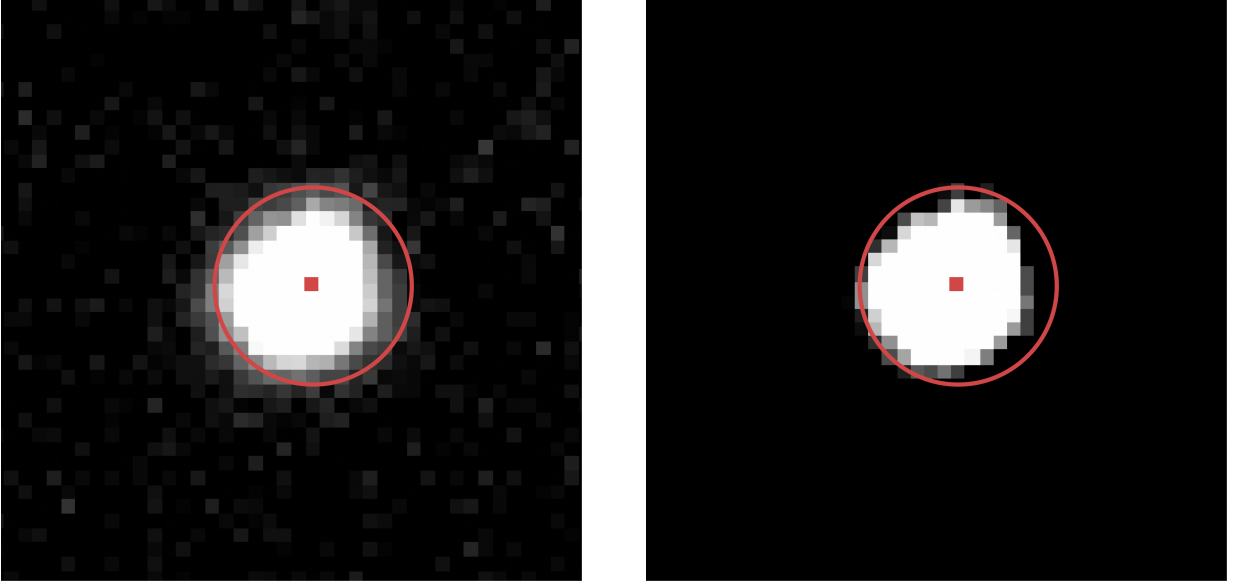
We performed this using `aperture_photometry` from the `photutils` library, which sums the flux within a given aperture. For each star, we utilized a circular aperture with a radius of 6 pixels. A radius of 6 pixels is sufficient as it was found to be enough to fit the brightest stars. Additionally, because the background flux around each star was found to be relatively uniform, we ran the aperture photometry with a global background subtraction rather than a local subtraction for each individual star. The global background that was subtracted was the median count value for the B and V frame respectively. This step also ensures that the common radius of 6 pixels is not overkill for the smaller point sources in the frame as the background flux is subtracted anyway.

To calculate the uncertainty from our measurement, we know that for an object, the measurement uncertainty comes from the shot noise of the object and the background noise

$$\sigma_{total} = \sqrt{\sigma_{obj}^2 + \sigma_{bkg}^2} = \sqrt{N_{obj} + \sigma_{bkg}^2} \quad (2)$$

where  $\sigma_{bkg}$  is the standard deviation of the pixels of the framer multiplied by the amount of pixels that are summed, which is equal to the area of the aperture (3).

$$\pi r^2 = \pi(6px)^2 = \pi(36px^2) = 113.1px^2 \quad (3)$$



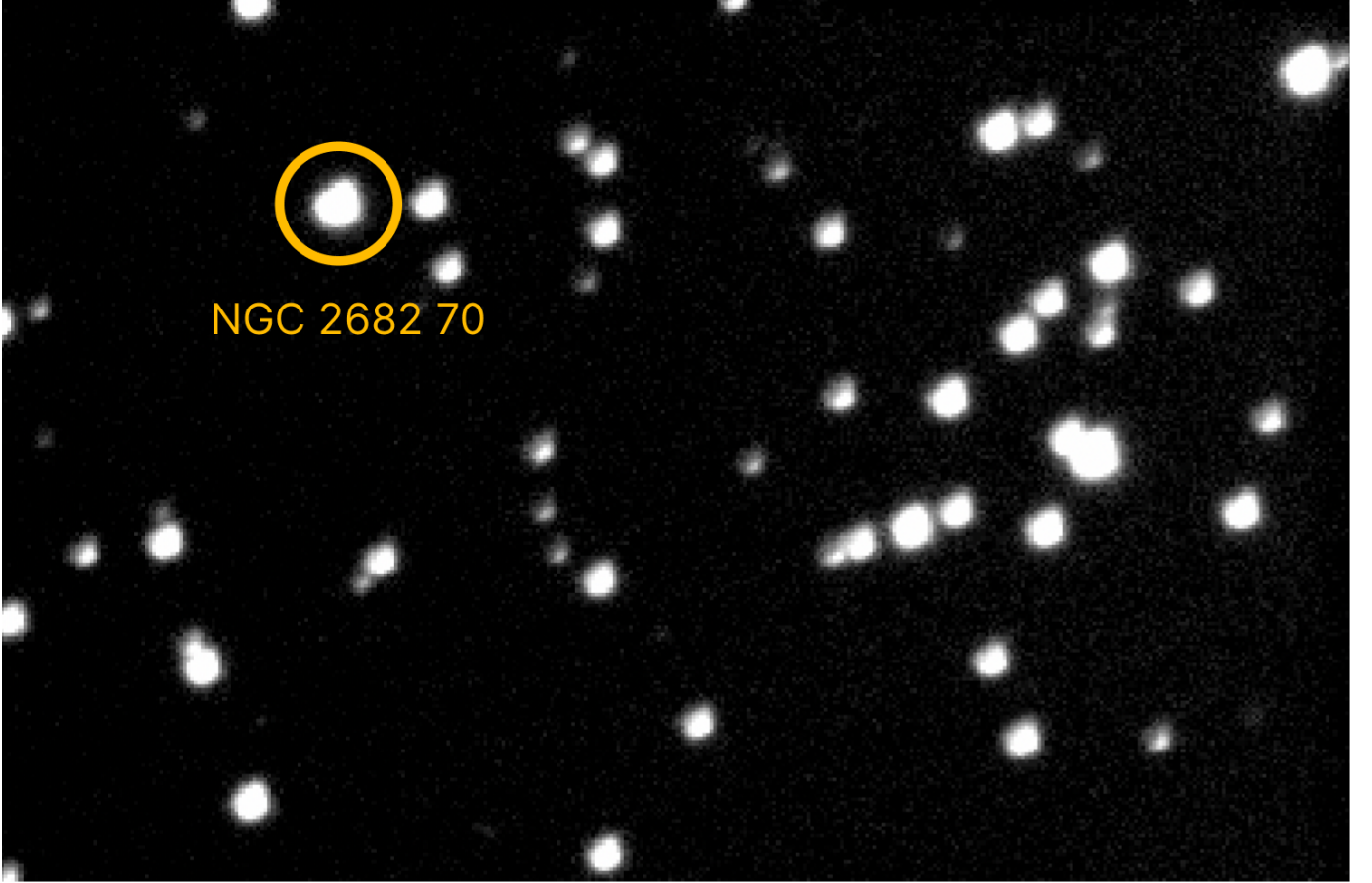
**Figure 2.** Visual representation of a circular aperture with background flux (left) and global background flux subtracted (right).

This process was repeated for both the M67 and SA101 region frames.

### 3.6. Magnitude Calibration

Before the flux can be plotted on the color magnitude diagram, it must first be converted from flux in units of pixel counts to instrumental magnitude. From there, it must be converted from instrumental magnitude to absolute magnitude, which are typically plotted on CMDs and are the values typically provided for magnitude and color by isochrones.

Initially, we intended to use a star in the SA101 region as a reference star with known values in the B and V magnitudes from the Landolt standard star observations (Landolt (1992)). These values can then be used to calibrate our flux counts to the actual magnitudes for our stars in M67. Fortunately, we discovered that there existed a star within our final images' field of view, NGC 2682 70, that has literature values for V magnitude and B-V.



**Figure 3.** Preview of the final image in the B filter and the identified star of documented B and V magnitude, NGC 2682 70.

This means that rather than basing our calibration on a relationship between flux and magnitude from a star in a different region entirely, we could calibrate the magnitude of all the stars in respect to a star within the same region. This is beneficial and preferred over using one of the stars in SA101 for two major reasons. The first being that part of the purpose of a reference star is to correct for atmospheric variability. By choosing a star within M67, we ensure that there is relatively little to no difference in airmass between stars in our observation. The second major reason we chose NGC 2682 70 was that the stars we captured in SA101 were of high magnitude (16-17), meaning that they were especially dim. Within the same exposure time as M67 of 45 seconds, we had obtained counts for the stars in SA101 that were only in the order of hundreds rather than thousands, like NGC 2682 70 has, meaning that they are more susceptible to noise. So we relate the observed flux and known magnitude of NGC 2682 70 and the observed flux of every other star in the region, through the magnitude relation.

$$m_1 - m_2 = -2.5 \log_{10} \left( \frac{F_1}{F_2} \right) \quad (4)$$



We can rearrange this relation to get an unknown magnitude of an astronomical object,  $m_1$ , from it's flux  $F_1$  and a known magnitude and flux  $m_2$  and  $F_2$ .

$$m_1 = -2.5 \log_{10}\left(\frac{F_1}{F_2}\right) + m_2 \quad (5)$$

We apply this equation to our list of summed flux for each star in terms of counts to our known reference star of NGC 2682 70, which has a V magnitude of 11.550 and B magnitude of 11.96 (Geller et al. (2015)). By repeating this process with the flux counts of NGC 2682 70 across the B filter and the V filter, we now have a value for the magnitude in each filter for every star. We must also account for the uncertainty. Since the value is the result of the ratio between the fluxes, which are two measured quantities, and multiplication by constant values, we have to adjust for the new combined uncertainty. When multiplying or dividing two measured quantities, the new uncertainty is described by the formula

$$\frac{\delta_z}{z} = \sqrt{\left(\frac{\delta_a}{A}\right)^2 + \left(\frac{\delta_b}{B}\right)^2} \quad (6)$$

Additionally, the uncertainty of the value,  $z$ , following a base 10 logarithm is approximately equal to  $0.434 \frac{\delta_z}{z}$ . The ratio is then also multiplied by -2.5, therefore the uncertainty must be equal to

$$|-2.5| \times 0.434 \frac{\delta_z}{z} = 1.085 \frac{\delta_z}{z} \quad (7)$$

Now, we adjust and convert each of the recorded values from apparent magnitude to absolute magnitude. The absolute magnitude is the same as the apparent magnitude, the observed energy flux, but if it were observed from a standardized distance of 10 parsecs away from the source. This denotes the intrinsic luminosity of a star and is the format of the values provided by our isochrones. The distance modulus (8) indicates the amount (in units of magnitudes) by which the distance from observer to source has dimmed the observed flux and if the distance to the source is known, can be used to convert our magnitudes to absolute magnitudes.

$$m - M = 5 \log_{10}\left(\frac{d}{10pc}\right) \quad (8)$$

Fortunately, we were also able to find an existing value of  $(m - M)_o = 9.70 \pm 0.05$  in literature for the bolometric distance modulus for M67 (Sarajedini et al. (2009)). We then subtract each of the apparent magnitudes by 9.70 in order to get the absolute magnitudes of each of the stars in M67. We can now easily generate the color values by subtracting the absolute V magnitudes from the B magnitudes to get  $B - V$ . To calculate the uncertainty for this  $B - V$ , we use the equation:

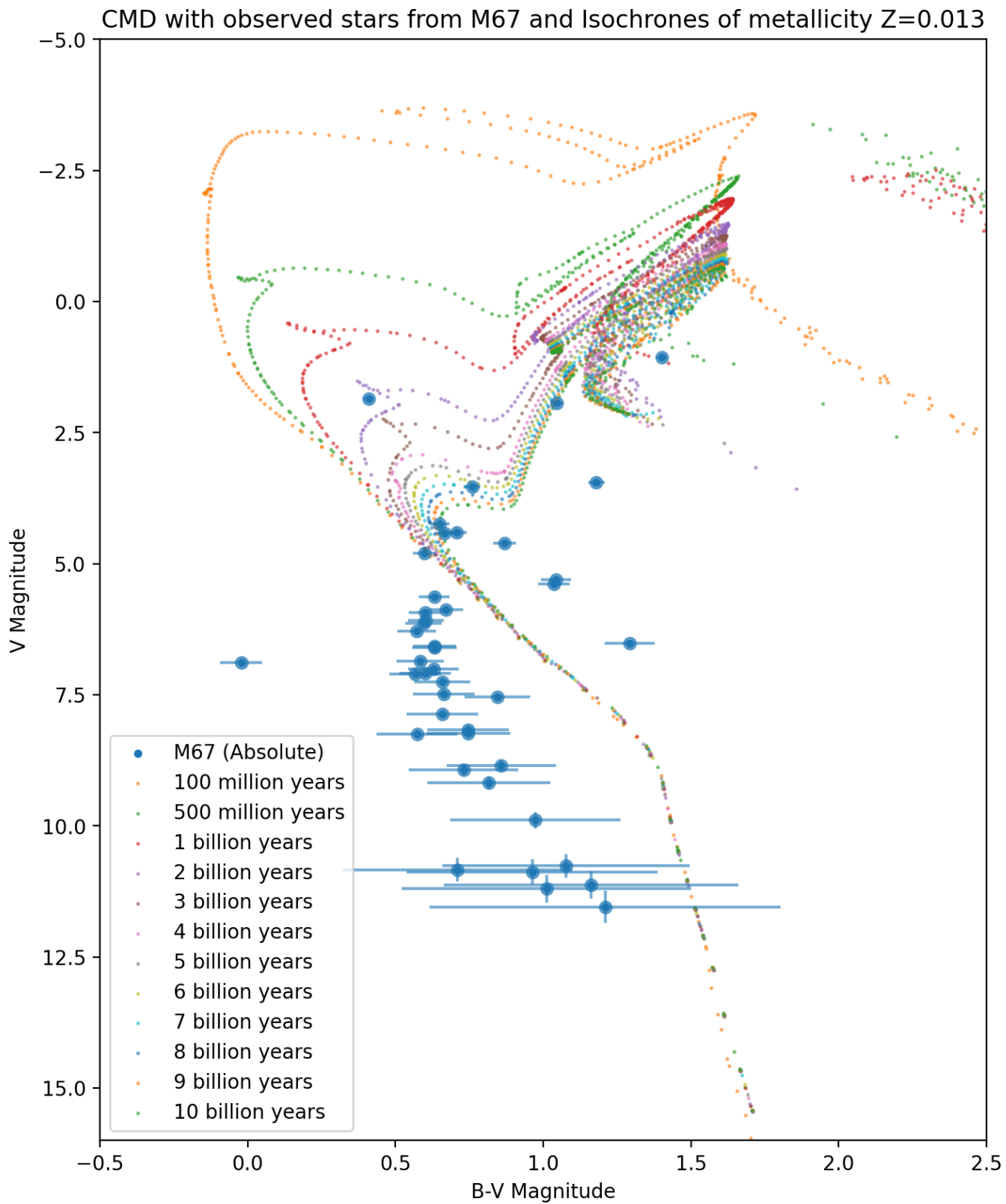
$$\delta_{B-V} = \sqrt{\delta_B^2 + \delta_V^2} \quad (9)$$

### 3.7. Generating and Plotting Isochrones

Before we plot the values we calculated for the observed stars in M67, we must obtain the isochrones across various ages. To generate the isochrones from the PARSEC stellar evolution code (Bressan et al. (2012)) there are other parameters for the model that must be decided. One of the main parameters that we must assume is the metallicity of the stars in the cluster. Since these stars are from an open cluster and are composed of the same material, the metallicity will be approximately the same between each of them. Furthermore, we found existing metallicity values of approximately  $Fe/H = 0.009 \pm 0.037$  for the stars that compose M67 in literature (Otto et al. (2026)). The PARSEC code accepts inputs in terms of  $Z$ , which is the fraction of metals in the sun's mass. This value in  $Fe/H$  represents the logarithmic ratio between iron and hydrogen relative to the Sun's value. Since 0 falls within the range of uncertainty for our metallicity value for M67, we can approximate the metallicity of M67 as a whole to be almost identical to the Sun's at  $Z = 0.013$  (Grevesse et al. (2012)).

## 4. RESULTS

The isochrones in figure 4 were generated with PARSEC version 1.2S, YBC + new Vega bolometric corrections, 0 interstellar extinction, and metallicity set to  $Z = 0.013$ .



**Figure 4.** A color magnitude diagram of our observed M67 magnitudes and a series of stellar isochrones generated from the PARSEC stellar tracks database.



In our color magnitude diagram, there is a clear visible region of observed stars clumped together in a line below where each of the isochrones begin. This region can likely be identified as the main sequence branch because of the density, and natural linear progression of stars of various masses. Additionally, there appears to be a clear main sequence turnoff point at around 5 V magnitude where the stars curve and spread out from that cluster.

While there is clearly a rough shape with evolutionary stages visible from the plotted observed color and magnitude, it does not completely align with our expected literature value of 4 billion years (Richer et al. (1998)). Our values are notably sitting shifted to the left, implying that there is a less vast difference between the B and the V magnitude than the stellar isochrones predict. Additionally, in the main sequence stars only the bottom most, dimmest stars fit with the stellar model as they have the largest uncertainty. Despite this we still are able to safely extract certain findings from our diagram, such as a lower bound. We can see that all of our plotted stars fit within the 1 billion year (red) isochrone's range and is likely confirmed to be older, considering no stars that were observed fall into the large tail that begins at roughly  $V = -2/5$  and  $B-V=2.3$  that is composed of 1 billion and 500 million year-old stars.

## 5. DISCUSSION

In this lab, we attempted to measure the magnitude of a number of stars within a region of open cluster M67 (NGC 2682) in order to derive an approximate value for M67's age. We attempted this by converting each stars' flux to magnitude by identifying a star with a known B and V magnitude and using the magnitude relation to calibrate it based on it's flux in terms of counts. After adjusting to get the absolute magnitude from the derived instrumental magnitude using the distance modulus, we were able to plot the the stars as a function of magnitude in the V filter and color (B-V magnitude) and compare it to a series of isochrones of different ages generated from the PARSEC stellar evolution library. Our initial proposal suggested observing stars from the Landolt standard star observations (Landolt (1992)) and using those known magnitudes to calibrate our data. Though we did observe SA101, we ultimately did not use that data as we fortunately were able to identify a star in our captured region of M67 with documented B and V magnitudes, NGC 2682 70. These values were preferred as they corresponded to a brighter magnitude and removed the impact of variation in airmass between the captured regions.

We found that using our data, it is possible to achieve at least a lower bound on the age of M67 of 1 billion years at the minimum. Though our data currently doesn't provide a clear upper bound that also clearly agrees with our expected literature value of 4 billion years, there is much room for improvement in future work. At the minimum, this lab can be repeated with more observations of M67 that span other portions of the region that weren't covered in our field of view. This would allow us to analyze more data with the same processing pipeline to identify whether there is an error with our analysis or if there was simply an insufficient sample space. Currently, there do already exist notable outliers, such as the star visible in figure 4 with  $B-V \approx 0.0$ . This can potentially be due to the inclusion of foreground or background stars that exist in our observations of the open cluster but don't actually belong to it, likely having a different age or chemical composition compared to the rest of the in-frame stars. Additionally, we can review each of our assumptions made from literature values to identify error with our analysis, with our data, or with those literature values themselves, including for the metallicity, distance modulus, magnitude, and the PARSEC stellar evolution library. Overall, while the quality of the results can likely be improved with the inclusion of further data, the lab was successful in identifying clear features in M67's evolutionary stages across it's shape and identifying a lower bound for it's age at the very least.

We would like to once again thank Dr. Simon Birrer and course TA Béla Arwen for always being on-call and providing us with constant guidance and answers to the queries we had across the entire semester. Additionally, we would like to thank the both of them for clear feedback on the presentation for this lab and the previous report. It was truly appreciated and will be utilized for any future work in the field.

## REFERENCES

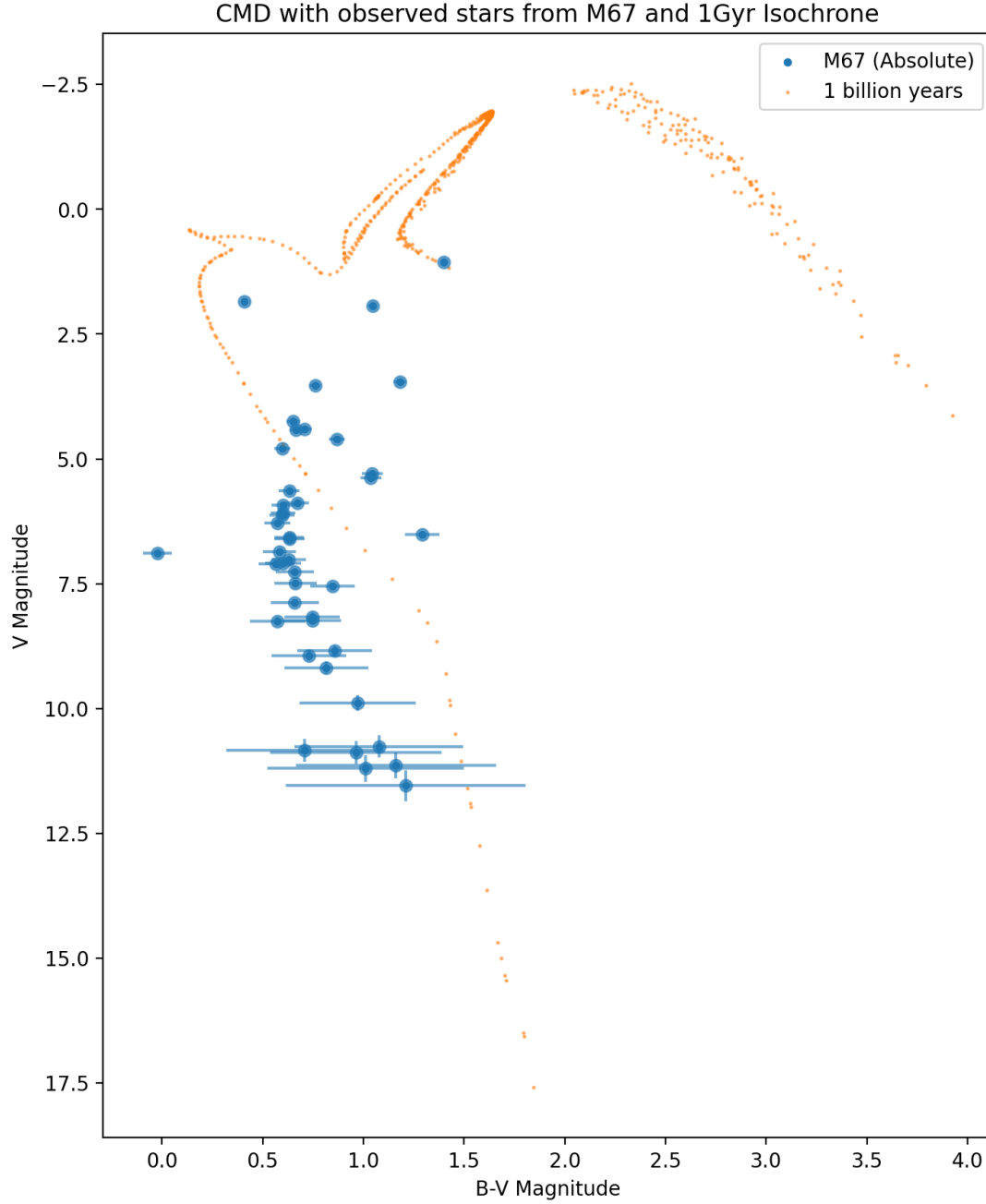
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| Bressan, A., Marigo, P., Girardi, L., et al. 2012, MNRAS,                                                      | Geller, A. M., Latham, D. W., & Mathieu, R. D. 2015, The                                 |
| 427, 127, doi: <a href="https://doi.org/10.1111/j.1365-2966.2012.21948.x">10.1111/j.1365-2966.2012.21948.x</a> | Astronomical Journal, 150, 97,                                                           |
|                                                                                                                | doi: <a href="https://doi.org/10.1088/0004-6256/150/3/97">10.1088/0004-6256/150/3/97</a> |

- Grevesse, N., Asplund, M., Sauval, A. J., & Scott, P. 2012, in *Astronomical Society of the Pacific Conference Series*, Vol. 462, *Progress in Solar/Stellar Physics with Helio- and Asteroseismology*, ed. H. Shibahashi, M. Takata, & A. E. Lynas-Gray, 41
- Landolt, A. U. 1992, *AJ*, 104, 340, doi: [10.1086/116242](https://doi.org/10.1086/116242)
- Otto, J. M., Frinchaboy, P. M., Myers, N. R., et al. 2026, *The Astronomical Journal*, 171, 91, doi: [10.3847/1538-3881/ae28d8](https://doi.org/10.3847/1538-3881/ae28d8)
- Richer, H. B., Fahlman, G. G., Rosvick, J., & Ibata, R. 1998, *The Astrophysical Journal*, 504, L91–L94, doi: [10.1086/311586](https://doi.org/10.1086/311586)
- Sarajedini, A., Dotter, A., & Kirkpatrick, A. 2009, *The Astrophysical Journal*, 698, 1872–1878, doi: [10.1088/0004-637x/698/2/1872](https://doi.org/10.1088/0004-637x/698/2/1872)
- Stetson, P. B. 1987, *PASP*, 99, 191, doi: [10.1086/131977](https://doi.org/10.1086/131977)

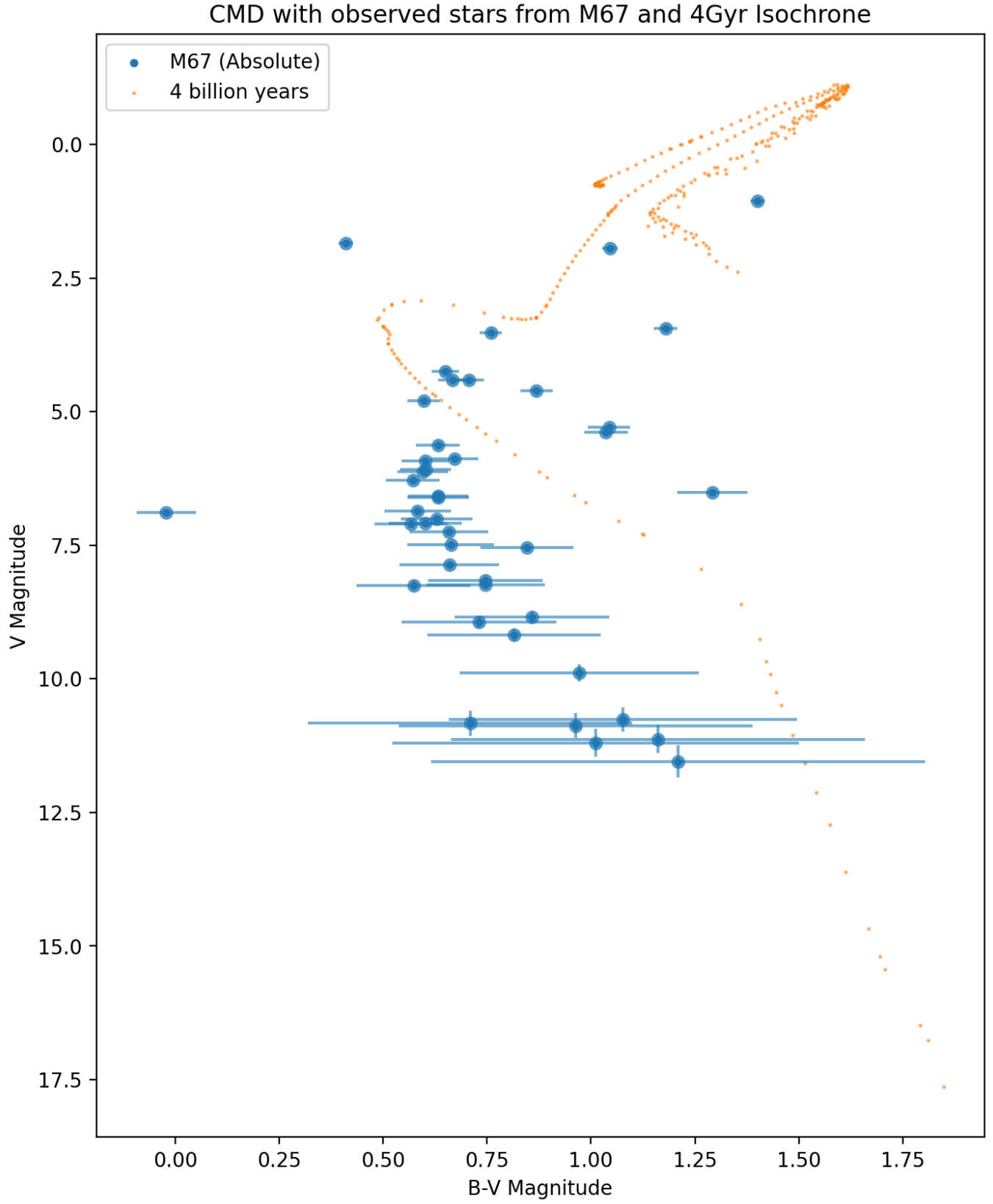
## 6. AI STATEMENT

No AI was utilized by the author in the completion of this lab- in both writing and completion of programming in the Jupyter notebook.

## 7. APPENDIX



**Figure 5.** Color Magnitude Diagram of observed stars from M67 and a isochrone of 1 billion year old stars with metallicity  $Z = 0.013$ .



**Figure 6.** Color Magnitude Diagram of observed stars from M67 and a isochrone of 4 billion year old stars with metallicity  $Z = 0.013$ .